Signal Harbour

Systematic multi-asset funds, a sector news-sentiment index, and a Streamlit research app

FINS3645 Project B
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Sample through December 2023

1. Customer, data, and backtest design

Signal Harbour is a research product for a small wealth-advisory firm. The day-to-day user is a portfolio analyst who needs comparable walk-forward evidence before an investment-committee discussion (Hutto and Gilbert, 2014, supply the baseline sentiment tool the course builds on). The menu covers Equity, Crypto, and Combined sleeves. Each sleeve is evaluated with equal weight, minimum variance, maximum Sharpe, and risk parity (Markowitz, 1952).

Prices are adjusted-close simple returns within ticker. Equity and Combined sleeves use a 252-session estimation window and 252-day annualisation. Crypto sleeves use a 365-day native window. Crypto returns are computed on the native calendar and only then left-aligned onto equity dates for Combined funds. That order keeps weekend compounding from being invented or deleted in the merge.

On monthly decision date T, target weights use returns strictly before T. Those targets become effective on the next session; the close-to-close return on T still uses prior holdings. Between rebalances, holdings drift with relative performance. Optimisation is long-only, with a 25% name cap in the broad sleeves. Covariance is annualised inside the solver. A candidate solution is kept only when it improves on equal weight.

Backtest conventions

The risk-free rate is zero in the Maximum Sharpe objective and in every reported Sharpe ratio. Headline fund results assume zero trading costs. Table 5 applies the same turnover-cost stresses to the MinVar base and the sentiment-fusion sleeve. The Streamlit application reads precomputed CSV files. It does not re-run VADER, rebuild the sentiment series, or execute backtests at runtime.

First-live timing

The decision date and the first earning date are not the same under this convention. After the 252-session warm-up, the first Equity and Combined decision date is 4 January 2021; those targets first earn a return on 5 January 2021. On the native crypto calendar, the first decision date is 1 January 2021 and the first live return is 2 January 2021. These are the starting dates of the out-of-sample paths in Figure A1, not the start of the raw price histories.

Why timing and drift are specified this way

A desk that forms weights after the close cannot book the full close-to-close return on T as if the new book had already been traded at the previous close. Assigning T to the old holdings and switching on the next session is the conservative sequence, and it is reproducible.

Drift is a separate issue. A strategy described as monthly rebalancing is not monthly if it is reset to target weights every day. Letting prices move the holdings, then measuring turnover against those drifted weights, keeps the wealth path and the cost exercise aligned. Annualising covariance inside the optimiser, and rejecting solutions that fail to beat equal weight, also responds to the project brief: SLSQP can report success without leaving its starting vector when daily variances are small. In the retained Equity sample, Minimum Variance differs from Equal Weight at every monthly decision. The mean half-L1 distance is about 0.72. That check shows the optimisation actually moved the weights.

2. Out-of-sample fund results

Table 1 and Figure A1 report the full fund menu. Among mixed-asset portfolios, Combined Risk Parity has the highest zero-risk-free-rate Sharpe ratio, 0.91. CAGR is 14.3%, volatility is 16.2%, and maximum drawdown is -19.3% (Figures A2 and A4). Combined Maximum Sharpe finishes with slightly more wealth (1.55x and a 15.7% CAGR), but its drawdown reaches -26.2%. These are research paths under the stated assumptions.

How to read the mixed-asset ranking

The mixed result is mainly about diversification under a calendar a desk can actually trade. A stand-alone crypto portfolio compounds through weekends because it is evaluated on 365 native days. In the Combined backtest, crypto returns are aligned to equity sessions, so the mixed desk earns that leg only when the equity book can also be traded. Under that constraint, Risk Parity repeatedly cuts exposure to the most volatile names in the trailing covariance estimate. It does so without forecasting expected returns.

Maximum Sharpe has a heavier estimation burden because it must estimate means as well as covariance. Over a three-year live window, noisy means encourage larger bets. That shows up in the deeper drawdown and in the concentrated December 2023 snapshot in Table 3. For a small advisory firm the distinction is practical. Combined Risk Parity is the stronger default comparison. Combined Maximum Sharpe belongs in a satellite discussion with clients who accept concentration and drawdown risk.

Equity and crypto sleeves

On equities, Equal Weight leads with a Sharpe ratio of 0.86. Risk Parity reaches 0.76, ahead of Minimum Variance at 0.47, but it still does not displace the transparent baseline. Optimisation is not pointless for that reason. Once the walk-forward rule is imposed, this sample did not pay enough for the extra estimation to beat Equal Weight. Minimum Variance does not forecast expected returns, so its weaker realised Sharpe is more plausibly linked to unstable covariance estimates and binding caps than to mean estimation.

Crypto Minimum Variance is the strongest crypto sleeve, with a 1.04 Sharpe ratio and a 62.1% CAGR. Crypto Maximum Sharpe is why CAGR and Sharpe are kept separate in this report: CAGR is -2.7%, even though the annualised arithmetic mean-to-volatility ratio remains positive at 0.35.

Table 1. Out-of-sample performance. CAGR is the annualised return from the growth-of-one path. Sharpe is annualised arithmetic mean divided by annualised volatility, risk-free rate zero. Equity-calendar funds use 252 days; native crypto funds use 365. Source: performance_metrics.csv. See Figures A1 and A4.

Fund	CAGR	Vol.	Sharpe	MaxDD	Terminal
Eq EW	13.3%	16.1%	0.86	-20.2%	1.45x
Eq MinVar	5.4%	12.7%	0.47	-14.9%	1.17x
Eq MaxSharpe	7.0%	17.5%	0.48	-22.8%	1.22x
Eq RP	10.4%	14.5%	0.76	-18.4%	1.34x
Cry EW	37.1%	81.7%	0.80	-81.7%	2.58x
Cry MinVar	62.1%	71.2%	1.04	-71.5%	4.25x
Cry MaxSharpe	-2.7%	76.2%	0.35	-85.6%	0.92x
Cry RP	41.0%	79.7%	0.83	-80.1%	2.80x
Comb EW	15.5%	21.5%	0.78	-27.7%	1.54x
Comb MinVar	5.4%	12.8%	0.48	-15.1%	1.17x
Comb MaxSharpe	15.7%	23.3%	0.74	-26.2%	1.55x
Comb RP	14.3%	16.2%	0.91	-19.3%	1.49x
Eq MinVar+Sent	5.3%	12.7%	0.47	-14.9%	1.17x

Concentration and the Combined sleeve

Figure A3 follows the decision weights of Combined Minimum Variance and Risk Parity, with an Other residual so that each area sums to one. Risk Parity stays more dispersed, which matches the shallower drawdown in Figure A2. Minimum Variance is willing to cluster in names that look low-volatility in the trailing covariance matrix. That is coherent with its objective. It can still become a problem when the selected cluster later weakens.

Across families, the Combined sleeve tempers crypto-native volatility while keeping exposure to both asset classes. Crypto Equal Weight and Crypto Risk Parity reach high terminal growth, but their drawdowns remain extreme. The -2.7% CAGR and positive 0.35 Sharpe for Crypto Maximum Sharpe also warn against reading one metric in isolation. Once the mixed fund is evaluated on the equity calendar, Combined Risk Parity, rather than the best pure-crypto result, is the more usable default for a mixed-client conversation.

Table 2. Latest Combined Risk Parity decision weights, top five. Source: latest_holdings_snapshot.csv.

Ticker	Weight
MRK	3.81%
ABBV	3.74%
WMT	3.37%
KO	3.19%
TMUS	3.17%

Table 3. Latest Combined Maximum Sharpe decision weights, top five. Source: latest_holdings_snapshot.csv.

Ticker	Weight
GE	25.00%
NVDA	20.09%
SO	16.86%
ADBE	10.32%
BTC-USD	10.29%

These tables are end-of-sample snapshots. They describe concentration at the final decision date, not the full path from 2021 to 2023. A large December 2023 weight cannot be used as an ex-post explanation for performance earlier in the sample.

3. Sentiment model, sector index, and fusion

Part B keeps the headline-processing rules from Part A: ticker-date-title duplicates are removed, and each headline is mapped to the same or next equity session. Base VADER and the Week 9 finVADER analyser are then re-scored without alteration for comparison. The Week 9 benchmark combines NLTK VADER, SentiBigNomics scaled by 0.1, and the Henry lexicon, matching the course helper and Korab's FinVADER resources. Loughran and McDonald (2011) give the wider rationale for finance-specific dictionaries.

Signal Harbour starts from that Week 9 base and adds documented masks, finance terms, and negation-aware phrases. The resulting score is multiplied by coverage confidence and by the elevated Attention Pulse confidence developed in Part A. Abnormally low headline volume never raises confidence. Missing ticker-days are inserted as neutral on a complete equity ticker-day grid before sector aggregation. The sector index in Figure A5 and the tradable signal therefore share the same no-news convention. Any score used in a portfolio is lagged by one equity session.

The sector index as an economic series

Figure A5 is a delayed description of the news climate. It is not a stand-alone timing rule. Context-weighted sector means are mildly positive over 2020 to 2023, but their ordering moves with the macro cycle. During the February to April 2020 COVID stress window, Energy is the weakest sector at about 0.005, while Technology and Real Estate are closer to 0.05. That pattern is consistent with an energy-demand shock and comparatively resilient digital and property-related language in the supplied corpus. It is not a statement about realised sector returns.

The ranking changes in 2022. Energy becomes the strongest sector on the index at roughly 0.061, while Materials and Communications sit near the bottom, around 0.023 to 0.030. A year dominated by inflation, energy cash flows, and supply concerns makes that ordering economically plausible. Healthcare rises from about 0.036 in 2021 to 0.053 in 2023. Technology recovers from 0.033 in 2022 to 0.051 in 2023.

Coverage is not uniform. Technology and Consumer average more than five headlines on news-bearing ticker-days; Materials and Real Estate are nearer two and have larger exact-zero raw-score shares (Figure A11). Context weighting therefore discounts sparse observations rather than treating every sector as equally well measured. Communications stays in a relatively narrow 0.027 to 0.038 range across 2020 to 2023. Energy moves from 0.017 in 2020 to 0.061 in 2022 before easing to 0.041 in 2023. Consumer remains roughly between 0.037 and 0.045. Neutral filling compresses the index level, so the useful information is mainly in relative movement across sectors and years. Figure A11 is the coverage check for whether a move reflects dense headline flow or a thin signal.

Continuity with Part A

Two design choices keep continuity with Part A. First, Attention Pulse remains a volume-confidence feature. It was descriptive in Part A and is not converted here into a contemporaneous trading trigger. Using only its elevated-volume confidence keeps the construction past-looking, without reversing the earlier claim that the pulse itself was not backtested alpha.

Second, the phrase and lexicon layer is recorded in the audit material rather than hidden inside a package label. The Week 9 FinVADER lexicons are vendored with Apache-2.0 notices. Signal Harbour’s own terms, phrases, and masks are identified separately. Without that record, calling the model “augmented finVADER” would say very little about what actually changed.

Distributional model comparison

Table 4 and Figure A8 compare polarity shares across a 4,000-headline sample. A lower neutral share is only a distributional observation. It is not evidence of better prediction. I labelled the 120-headline worksheet myself. Signal Harbour buckets agree with those labels on 60.8% of the full sample (56.0% development, 68.9% holdout; n=120). The automated pseudo-label field is not treated as ground truth. Figure A9 therefore looks at component effects through ablation on the Week 9 analyser rather than reporting labelled precision or recall. On the 800-headline ablation sample, removing neutral overrides has no effect: *no_neutral_overrides* matches *full_signal_harbour* because the retained masks rarely fire. The components that visibly move the neutral share are custom terms and phrases. Figure A10 reports active ticker counts. Figure A11 is the separate coverage-intensity exhibit.

Table 4. Model polarity shares on 4,000 headlines. The comparison is distributional only. Source: sentiment_model_comparison.csv; Figure A8.

Model	Mean	Positive	Neutral	Negative
Base VADER	0.106	37.5%	49.6%	12.9%
Week 9 finVADER	0.069	26.6%	62.1%	11.3%
Signal Harbour	0.081	29.8%	58.7%	11.6%

Fusion result

The main fusion test applies the lagged context-weighted score to Equity Minimum Variance. Figure A6 shows that the two wealth paths are almost indistinguishable, but the overlay is slightly worse: the zero-risk-free-rate Sharpe ratio falls from 0.4740 to 0.4711 and CAGR from 5.36% to 5.32%. Figure A7 and Table 6 extend the stress across tilt strengths and base portfolios. None of the tested specifications overtakes Equity Equal Weight. Table 5 then applies the same turnover-cost assumptions to the MinVar base and the fusion sleeve. Even modest costs preserve, and slightly widen, the disadvantage of the overlay. What remains usable is a continuous, look-ahead-safe monitoring pipeline. It is not a marketed alpha claim.

Table 5. Base versus fusion under equal turnover costs. TO is average annual turnover. Source: fusion_cost_sensitivity.csv.

Sleeve	Cost	Sharpe	CAGR	TO	Terminal
Base	0 bp	0.474	5.36%	1.90	1.169x
Fusion	0 bp	0.471	5.32%	1.91	1.167x
Base	5 bp	0.467	5.26%	1.90	1.165x
Fusion	5 bp	0.464	5.22%	1.91	1.164x
Base	10 bp	0.459	5.16%	1.90	1.162x
Fusion	10 bp	0.456	5.12%	1.91	1.161x
Base	25 bp	0.437	4.86%	1.90	1.152x
Fusion	25 bp	0.434	4.82%	1.91	1.151x

Table 6. Fusion robustness across equity bases and tilt strengths at zero cost. Source: fusion_sensitivity.csv; Figure A7.

Base	Tilt	CAGR	Sharpe	MaxDD
EW	0.15	13.24%	0.853	-20.2%
EW	0.35	13.17%	0.849	-20.2%
EW	0.60	13.08%	0.845	-20.3%
RP	0.15	10.39%	0.753	-18.4%
RP	0.35	10.34%	0.751	-18.4%
RP	0.60	10.29%	0.748	-18.5%
MinVar	0.15	5.34%	0.473	-14.9%
MinVar	0.35	5.32%	0.471	-14.9%
MinVar	0.60	5.29%	0.469	-14.9%

The negative fusion result does not mean finance-aware sentiment has no research value. Table 4 and Figures A8 and A9 show that Week 9 finVADER and Signal Harbour redistribute polarity mass relative to social-media VADER, and the Explain view can identify the terms and confidence weights behind a lagged score. The narrower conclusion is that, in this 2021 to 2023 equity sample, a multiplicative portfolio tilt was not rewarded after the one-session lag. That outcome is consistent with headline tone being partly contemporaneous with price information already present in the estimation window, and with the lag removing the most mechanical portion of any apparent edge. Repeating the test on Equal Weight and Risk Parity bases also shows that the result is not merely an artefact of a damaged MinVar path. Recommendation 3 follows from that: keep sentiment for monitoring and explanation, and do not present it as an independent return engine.

4. The Signal Harbour application

The application follows the analyst's pre-committee workflow. Each tab answers a different question using precomputed research files, which keeps the free-tier deployment light and makes the displayed evidence reproducible.

Compare. Rank sleeves by zero-risk-free-rate Sharpe and terminal growth, then plot a shortlist of wealth paths from the research sample through December 2023.

Fact sheet. Inspect one fund at a time: CAGR, volatility, Sharpe, drawdown, and the latest decision weights. A residual is stated whenever only the top holdings are displayed.

Allocate. Combine completed fund return streams without re-optimising names. Equity-only blends use intersecting equity sessions; blends containing crypto use the native union calendar.

Sentiment and Explain. Review the sector news climate, model comparisons, coverage diagnostics, and the prior-session fields that produced the lagged tradable score. Headlines are not re-scored live.

Calendar treatment in Allocate

Allocate uses a different calendar rule from the Combined-fund backtest because it mixes already-built return streams rather than re-optimising the underlying names. Any selection containing a Crypto fund uses the native union calendar, annualises at 365 days, and records zero for a closed equity leg. Equity-only blends remain on intersecting equity sessions and use 252-day annualisation. A 60/40 equity-crypto research allocation can therefore retain crypto weekend returns instead of silently dropping Saturdays.

A few product choices are worth stating because they affect what a user actually sees. The light theme is pinned so a dark-mode browser cannot place white headings on a pale custom background. Allocate does not imply that a crypto sleeve exists only on equity-open dates. Explain displays the signal-date fields that entered the one-session lag; same-day headlines are not paired with yesterday’s tradable score. The fact sheet states the residual weight whenever the top-20 table is not the complete portfolio.

For the advisory firm described in Part A, the interface maps onto a short meeting sequence. Begin with Combined Risk Parity and its drawdown profile (Recommendation 1; Sharpe 0.91). Introduce Combined Maximum Sharpe only when a client asks for more growth and accepts a deeper loss profile (Recommendation 2). Open Explain when the committee wants news-context colour, with Recommendation 3 already setting the boundary: sentiment is a monitoring layer, not a trading engine. If Energy’s stronger 2022 news-climate reading is questioned, the answer is in Section 3. Every series is labelled as an out-of-sample research path through December 2023 rather than as a live production feed.

5. Recommendations

The three recommendations below are taken from the verified tables and exhibits.

1. Default mixed sleeve. Use Combined Risk Parity as the primary balanced multi-asset comparison within this sample. Its zero-risk-free-rate Sharpe ratio is 0.91 and its maximum drawdown is -19.3% (Table 1; Figures A1, A2 and A4).

2. Satellite growth sleeve. Keep Combined Maximum Sharpe as an optional higher-growth alternative rather than the default. Terminal wealth reaches 1.55x, but maximum drawdown deepens to -26.2% and the latest holdings are more concentrated (Tables 1 and 3; Figure A3).

3. Sentiment as monitoring, not alpha. Retain the lagged context-weighted layer in the explainability workflow, but do not market it as a return engine. Relative to Equity MinVar, Sharpe moves from 0.4740 to 0.4711 and deteriorates further under equal cost assumptions (Figure A6; Tables 5 and 6).

6. Critical reflection and limits

Several choices make the study internally consistent and, at the same time, narrow what can be claimed from it.

Short live window. Equity and Combined returns run from 5 January 2021 to December 2023. The rankings are therefore exposed to the COVID recovery, the 2022 inflation and rate shock, and the 2023 equity rebound. A longer sample could reorder Equal Weight and Risk Parity.

Headline results exclude costs. Table 1 is a zero-cost laboratory. Table 5 shows that even 5 to 25 basis points per unit of turnover reduce the fusion result. Applying the same grid to every fund would probably narrow the gap between higher-turnover Maximum Sharpe paths and lower-turnover Risk Parity.

Uneven news coverage. Materials and Real Estate have thinner headline intensity and larger exact-zero raw-score shares than Technology or Consumer (Figure A11). Their sector readings require greater caution.

Labelled sentiment evidence is a small worksheet, not a precision claim. I labelled all 120 headlines in *kaiyuan_review_label*. Agreement with Signal Harbour buckets is 60.8% on that sheet (headline_kaiyuan_label_agreement.csv). That is a polarity-match rate on n=120, not labelled precision or recall.

End-point holdings are not path explanations. The December 2023 snapshots in Tables 2 and 3 describe concentration at the last decision date. They cannot explain the complete 2021 to 2023 wealth path.

Product-level reflection

Building Signal Harbour on the actual Week 9 analyser was a stronger continuity choice than attaching a few finance terms to plain VADER. The report records the provenance of both the course lexicons and the project-specific additions, and the application is explicit that it reads precomputed files rather than re-scoring headlines at runtime. Under those conditions, credibility comes from design quality, auditability, and restrained claims, not from the appearance of a live model.

If I extended the project, I would make three changes. First, I would apply the turnover-cost grid from Table 5 to every fund family so that Recommendation 1 is tested under the same frictions. Second, I would expand the labelled worksheet beyond n=120 before treating polarity agreement as a model-selection statistic. Third, I would publish the public GitHub repository and Streamlit deployment, then resolve any remaining theme issues before submission. Those steps would improve validation and delivery. They would not alter the central result that the sentiment overlay is economically flat to negative in this sample.

What out-of-sample means here

The fund weights are walk-forward and use only past returns, and the tradable sentiment input is delayed by one equity session. The research choices around lexicon retention, tilt strength, and the base sleeve receiving the overlay were nevertheless made with knowledge of the full study period. That is a normal limitation in a fixed-sample coursework project. The negative fusion result should therefore be read as a stress test of one pre-specified overlay design, not as proof that every possible news rule failed under a sealed holdout. The development-holdout split in the headline worksheet offers partial discipline for labelling work, but it is not a complete untouched test set.

Signal Harbour gives a small advisory desk a coherent menu of systematic funds, a lagged sector news-climate monitor linked to Part A, and an auditable explanation for why a name was tilted. It should not claim that the tilt paid in this sample, and it should not hide the calendar, cost, or coverage frictions that an investment committee would reasonably raise. Remaining packaging work is operational: keep the light theme pinned and verify that the live URL matches the version described here.

Appendix A. Exhibits

Each exhibit is discussed in Sections 2 to 4. Captions restate the sample, source, and the claim the figure is meant to support.

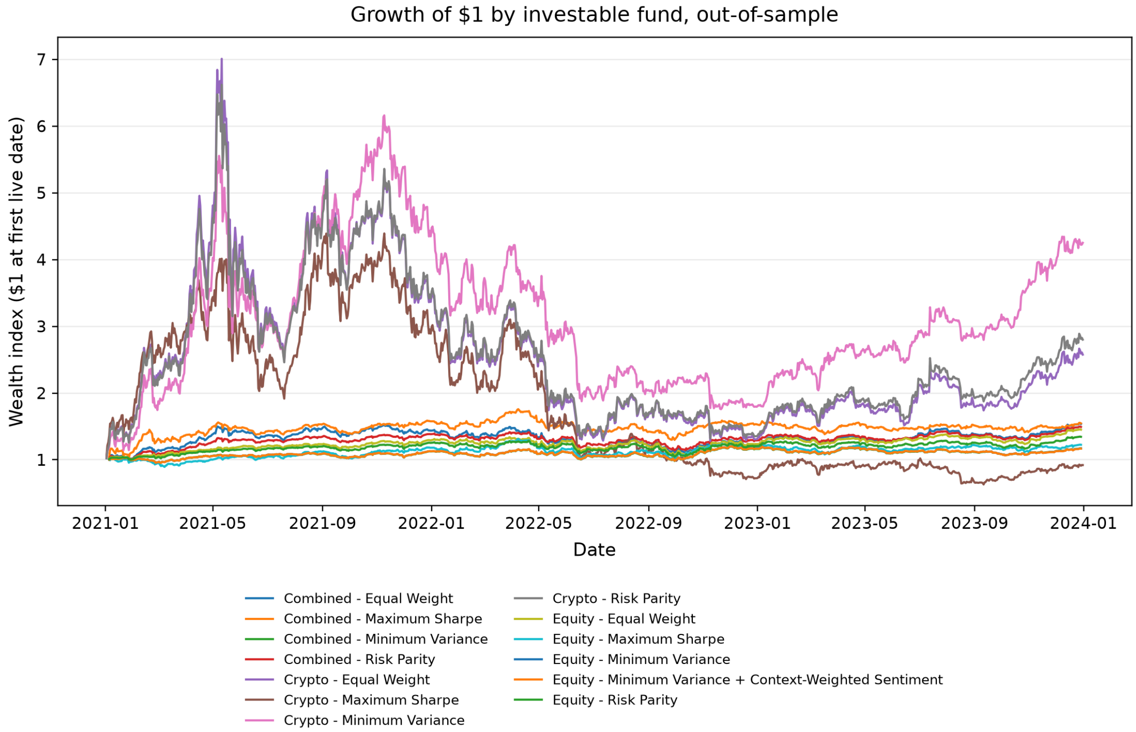

Figure A1. Growth of $1 across funds, out-of-sample through December 2023. Combined Risk Parity is steadier than Combined Maximum Sharpe; Crypto Minimum Variance leads terminal wealth among the crypto sleeves. Source: fund_returns.csv.

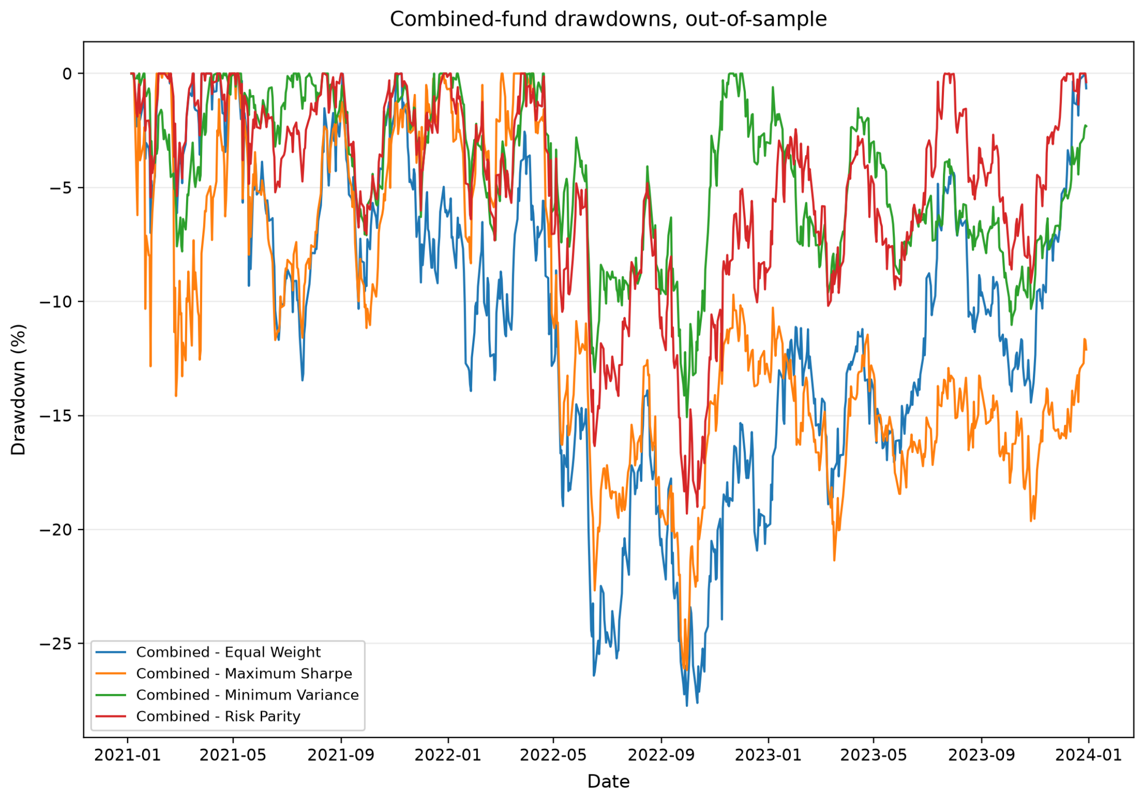

Figure A2. Combined-fund drawdowns. Risk Parity has the shallower path, supporting Recommendation 1; Maximum Sharpe reaches the deeper trough expected of the higher-growth satellite. Wealth paths prepend 1.0 so that the first loss is visible.

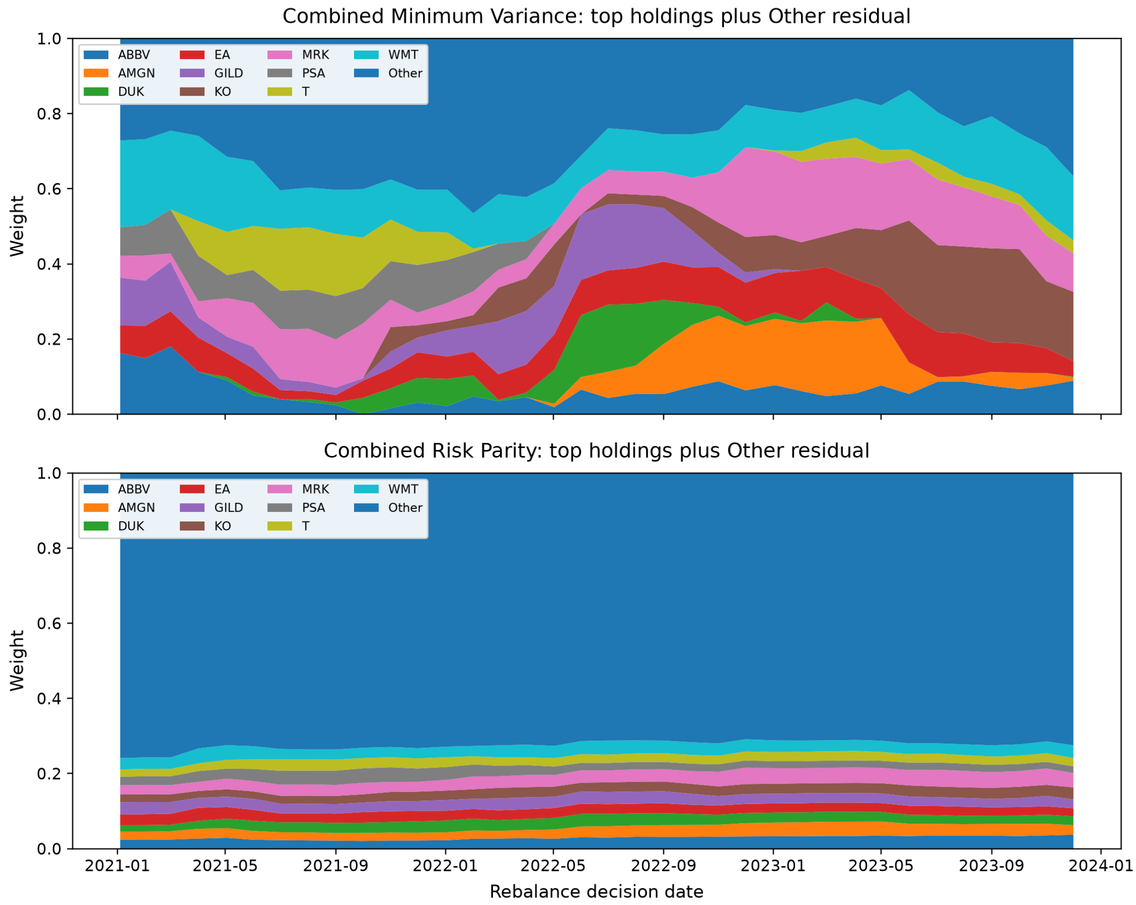

Figure A3. Combined Minimum Variance and Risk Parity decision weights, with an Other residual so the stacked areas sum to one. Maximum Sharpe concentration is summarised in Table 3 rather than in this panel.

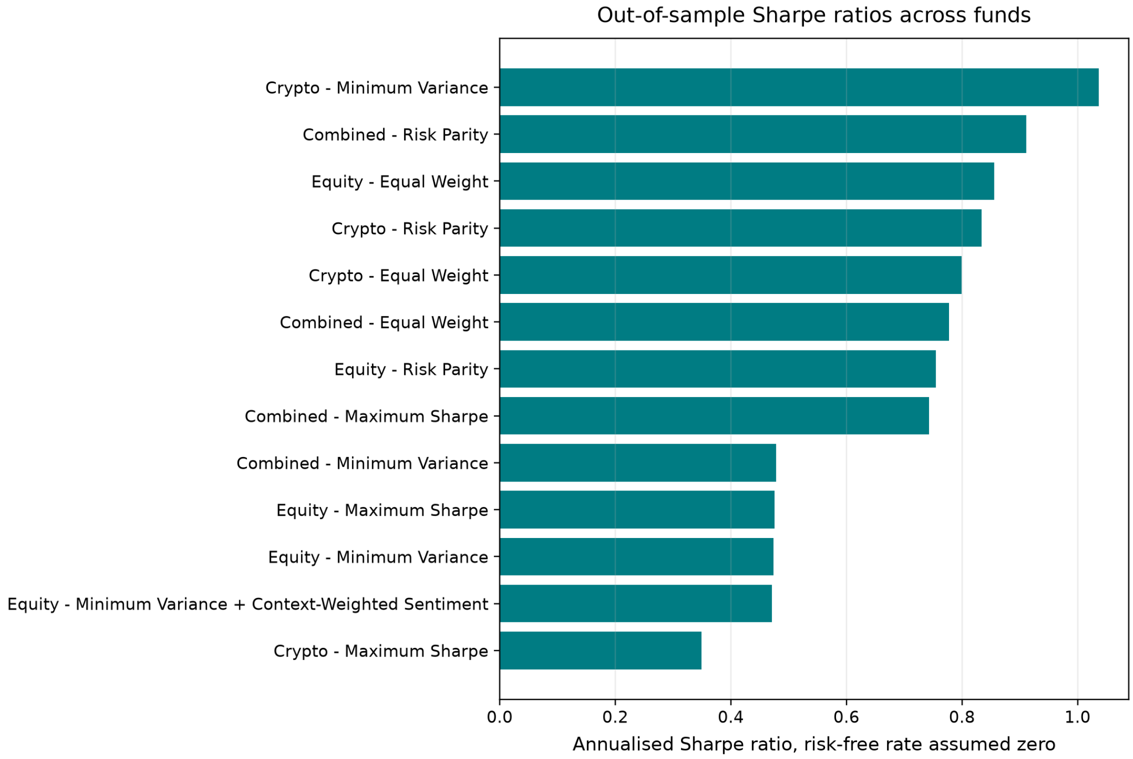

Figure A4. Zero-risk-free-rate Sharpe rankings. Combined Risk Parity leads the mixed sleeves; Equity Equal Weight leads equities; Crypto Minimum Variance leads crypto. Annualisation is 252 or 365 days by sleeve.

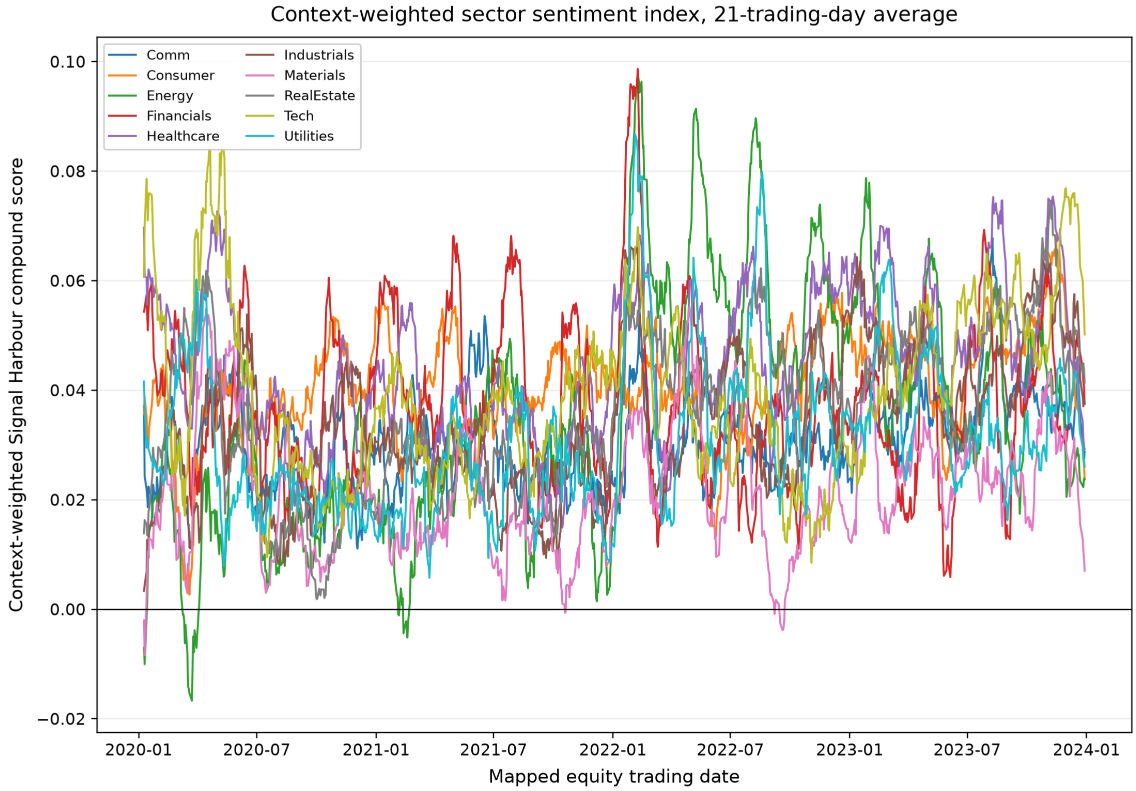

Figure A5. Context-weighted sector sentiment, shown as a 21-session average. The series uses the complete ticker-day grid, sets no-news observations to neutral, and is lagged before any trading use.

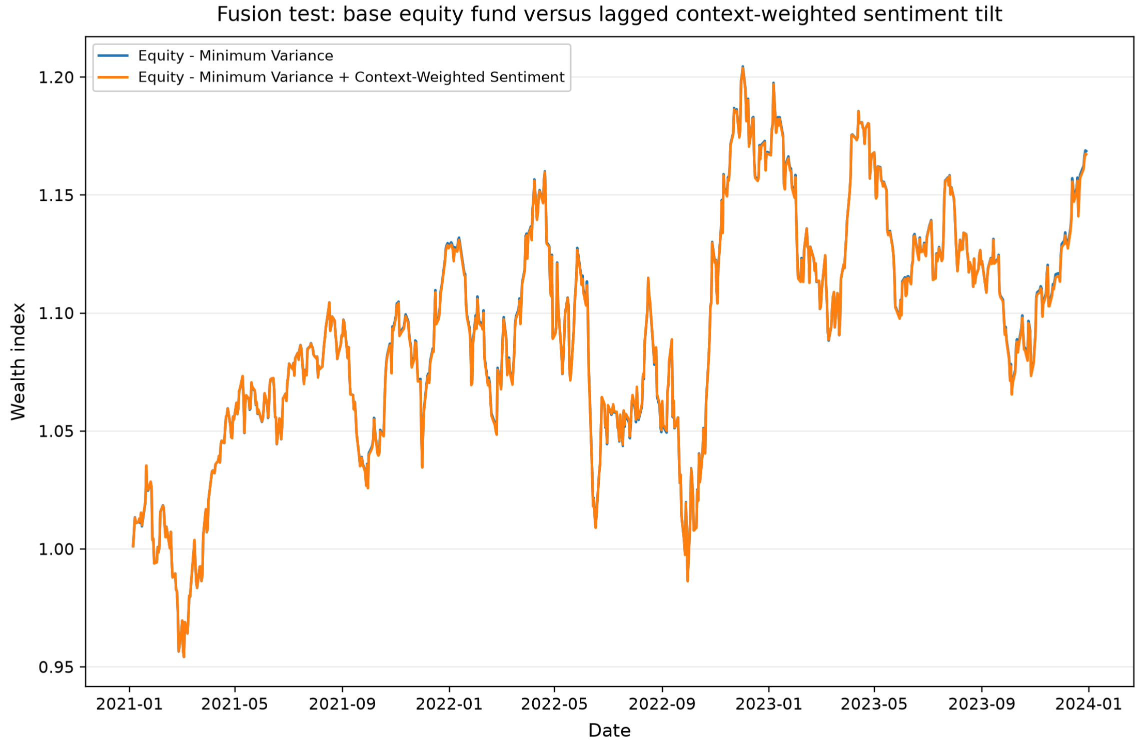

Figure A6. Equity Minimum Variance against the lagged context-weighted sentiment tilt. The overlay does not improve the corrected baseline, which supports Recommendation 3.

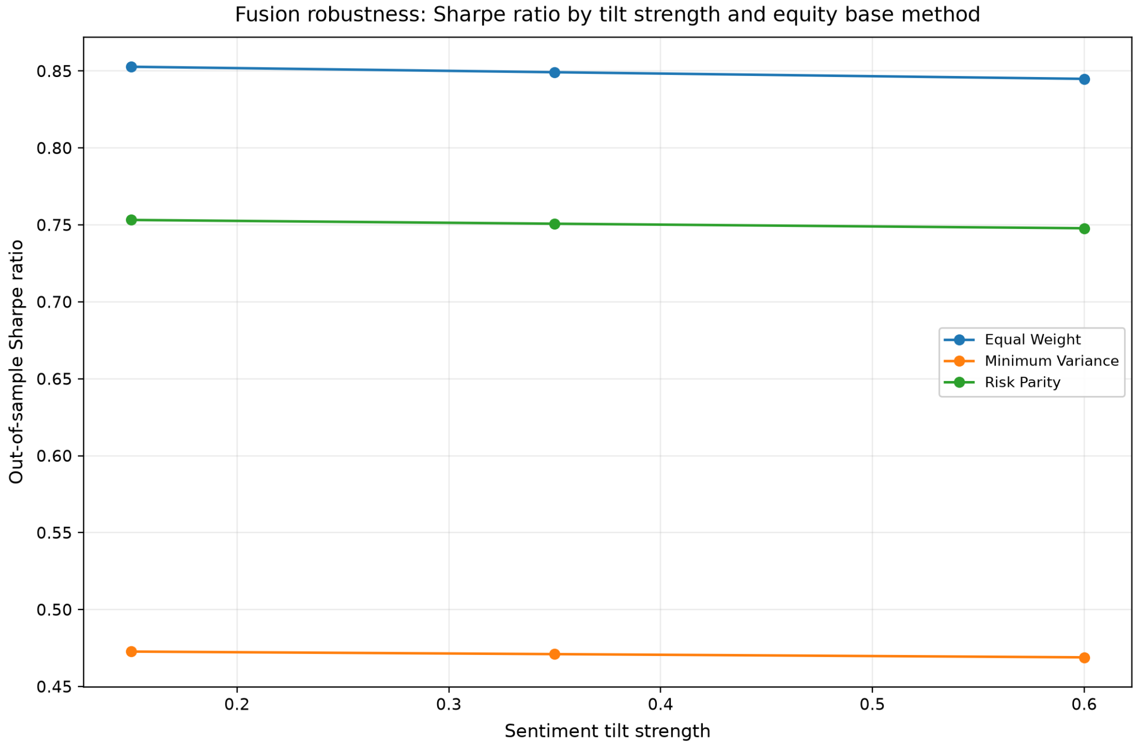

Figure A7. Fusion sensitivity by tilt strength and equity base at zero cost. None of the tested specifications overtakes Equity Equal Weight (Table 6).

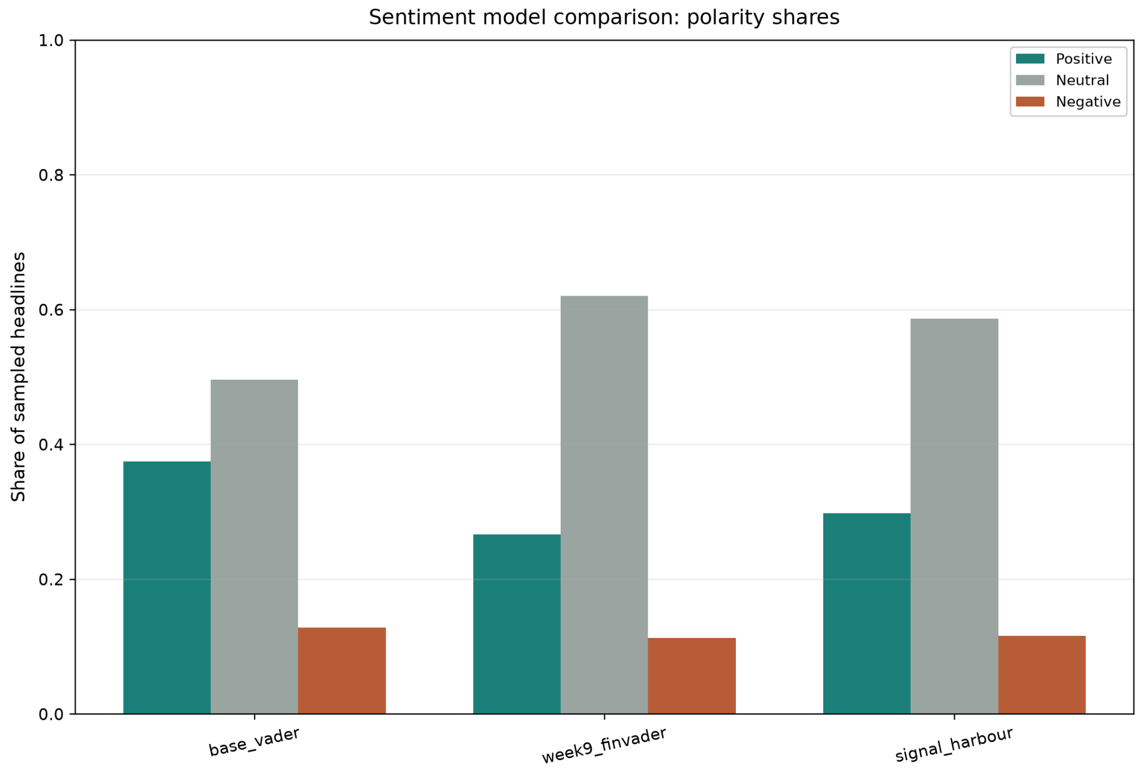

Figure A8. Polarity shares for Base VADER, Week 9 finVADER, and Signal Harbour on 4,000 headlines (Table 4). The comparison is distributional only.

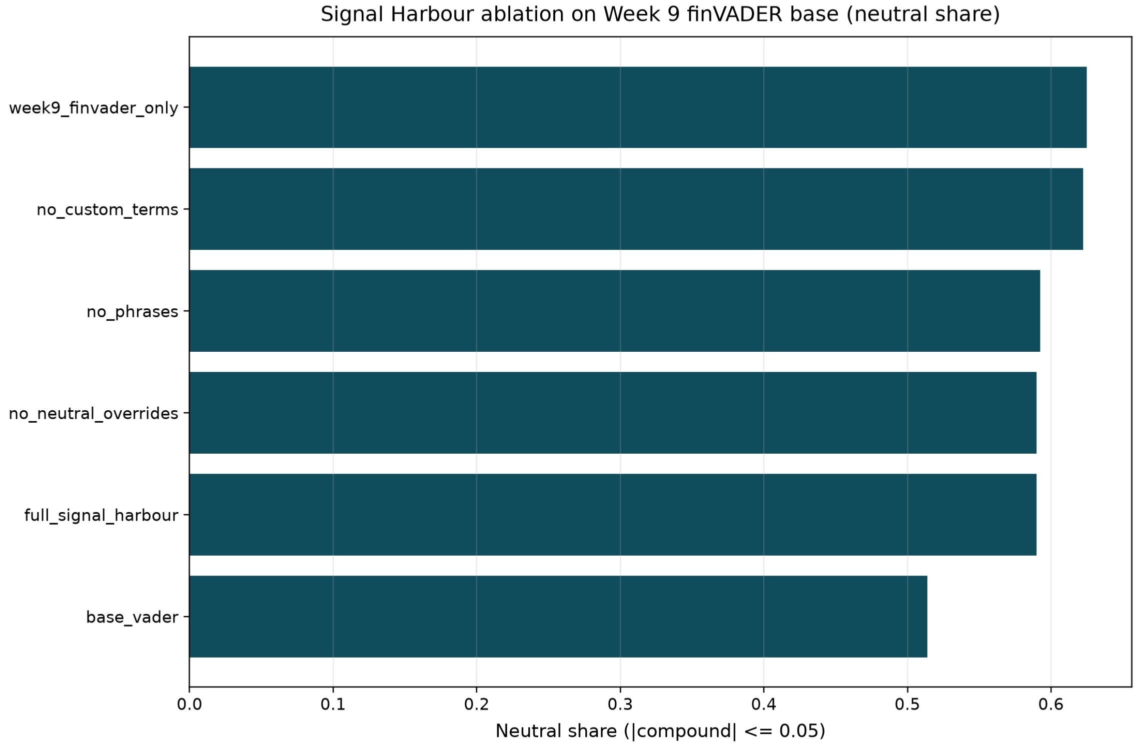

Figure A9. Ablation of Signal Harbour components on the Week 9 finVADER base analyser, with Week 9-only and plain-VADER references.

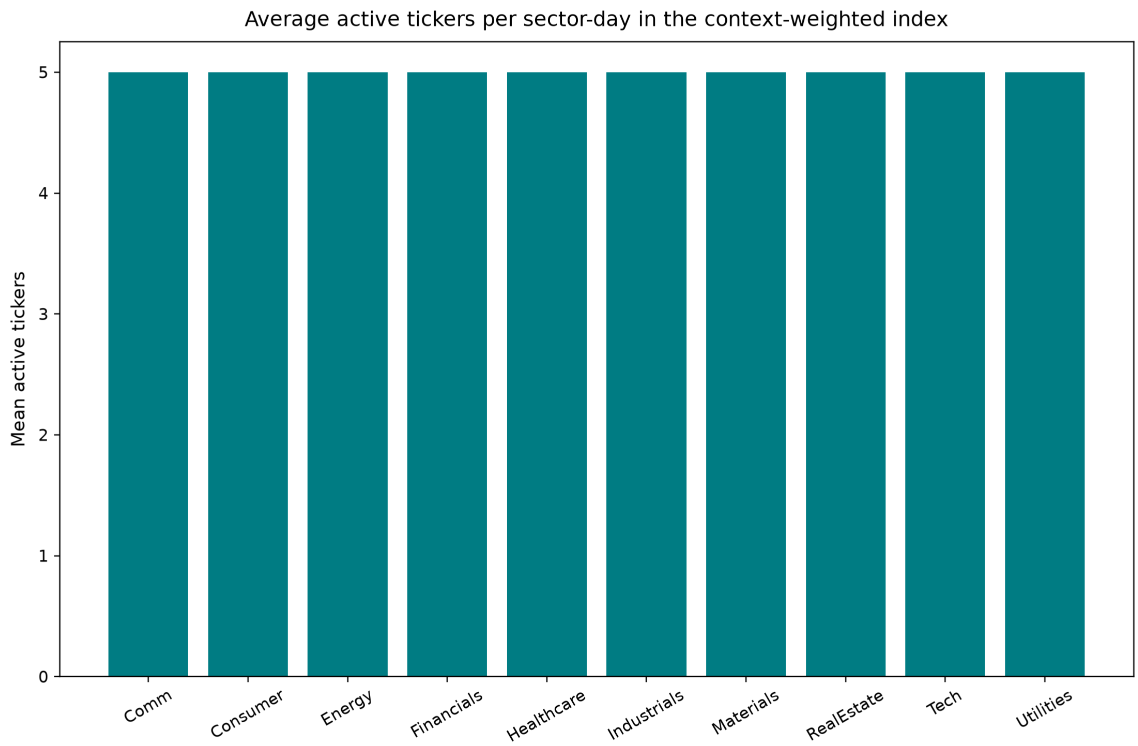

Figure A10. Average active tickers per sector-day under the complete-grid context-weighted index. This is the coverage companion to Figure A5.

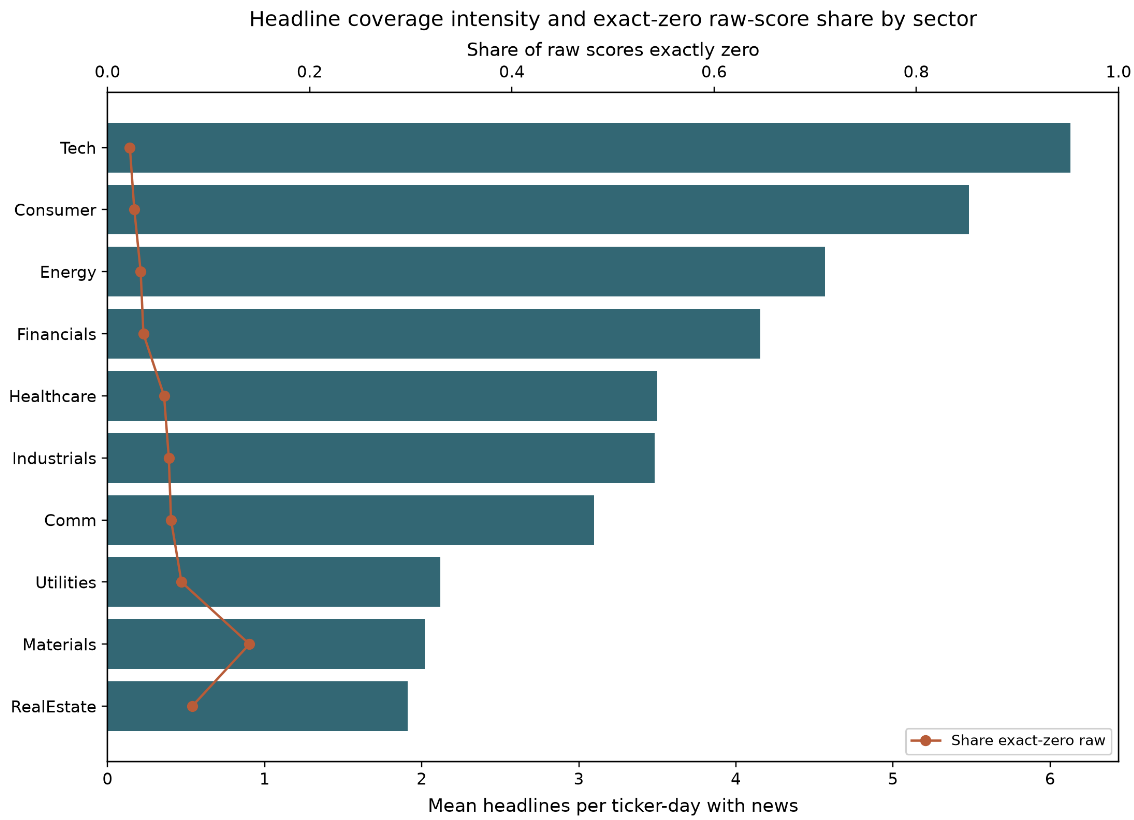

Figure A11. Sector coverage intensity: mean headlines on news-bearing ticker-days (bars) and the share of exact-zero raw scores (line). This exhibit is distinct from the polarity comparison in Figure A8.

Appendix B. Core equations and references

Minimum variance

minimise w^T SigmaAnn w subject to 1^T w = 1 and 0 <= w(i) <= wMax

SigmaAnn = ppy * Cov(R), where ppy is 252 on the equity calendar or 365 on the native crypto calendar.

Maximum Sharpe

maximise (w^T muAnn) / sqrt(w^T SigmaAnn w)

The risk-free rate is zero; the long-only and name-cap constraints are unchanged.

Reported performance

Sharpe = (mean(r) * ppy) / (std(r) * sqrt(ppy))

CAGR = WT raised to the power (ppy/T), minus 1

Context-weighted score

sCw(i, t) = s(i, t) * coverageConf(i, t) * attentionConf(i, t)

Tradable score = sCw(i, t minus 1).

Sentiment tilt

wTilt is proportional to w * exp(lambda * (score minus mean(score))). The tilted vector is then projected back onto the capped simplex.

References

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